

Direction of arrival tracking using adaptive robust subspace decomposition and Kalman filter

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Abstract—In this paper, we consider directions of arrival estimation and tracking in a noisy and relatively adverse scenario. We thus use a robust subspace tracking method called HTFAPI that exploits a weighted least squares criterion to enhance robustness, followed by a smoothing Kalman filter to improve the estimation. Simulated experiments are then presented to show that the proposed solution enhances the performance when dealing with corrupted data in an impulsive noise environment.

Index Terms—Robust subspace tracking, Directions of arrival, Kalman filter, Impulsive noise

I. INTRODUCTION

Directions of Arrival (DoA) estimation and tracking have long presented a great interest in signal processing. In the array signal processing literature, we can come across several research axes on this subject. The high-resolution methods that appeared in that context were the MULTiple Signal Classification (MUSIC) method [1] and the Estimation of Signal Parameters via Rotational Invariance Techniques (ESPRIT) algorithm [2]. However, these methods are very expensive in terms of complexity due to the eigendecomposition of the data covariance. Thankfully, adaptive algorithms with linear complexity have emerged to lower the cost of this eigendecomposition step, such as the Projection Approximation Subspace Tracking (PAST) [3], Orthogonal PAST (OPAST) [4], PAST with the deflation technique (PASTd) [3], Fast Approximated Power Iteration (FAPI) [5] and Fast Data Projection Method (FDPM) [6]. Furthermore, to improve the results of these algorithms, several works have proposed to add a Kalman filter to smooth the tracking of the DoA [7]–[9]. Despite their efficiency, these algorithms lack of robustness. Indeed, in the presence of impulsive noise, these algorithms diverge and the estimation error of the DoA becomes too large. In this context, one can find several subspace tracking algorithms that deal with

this problem [10]–[13]. It has been shown in [14] that the algorithm called Mahalanobis FAPI (MFAPI) and its lower cost version Hard Thresholding FAPI (HTFAPI) out-perform the other algorithms in the presence of data corrupted by impulsive noise or outliers.

The latter algorithms propose to replace the least squares (LS) criterion of the subspace estimation with a more appropriate weighted criterion WLS “Weighted LS” in order to mitigate the impact of corrupted data through the choice of a proper attenuation coefficient as in [10]. In addition, they use the projection approximation proposed in [5] which helps to reduce significantly the computational cost.

In this article, we propose to smooth the results obtained by the HTFAPI algorithm by using a Kalman filter. More precisely, the directions of arrival given by HTFAPI followed by ESPRIT algorithms are considered to be measurements for Kalman’s filter. Besides, the parameters of the latter are chosen dynamically according to the weighting coefficient of the algorithm HTFAPI.

The rest of this article is organized as follows. Section II details the problem statement. Then, the HTFAPI algorithm is introduced in section III. The proposed algorithm is presented in section IV, while the results of simulations are given in section V. Finally, section VI concludes this paper.

II. PROBLEM STATEMENT

We consider a uniformly spaced antenna array of n sensors receiving plane waves emitted by p narrow band source signals. We will denote $\mathbf{X} = [\mathbf{x}(1); \mathbf{x}(2); \dots; \mathbf{x}(N)]$ a sequence received in “ N ” time units where $\mathbf{x}(t)$ is the vector of observation of size n at time “ t ” given by:

$$\mathbf{x}(t) = \sum_{k=1}^p s_k(t) \mathbf{a}(\omega_k) + \mathbf{n}(t) = \mathbf{A} \mathbf{s}(t) + \mathbf{n}(t) \quad (1)$$

where:

- $\mathbf{s}(t)$: is a random signal vector corresponding to the data emitted by each source with the covariance $\mathbf{C}_s = E[\mathbf{s}(t)\mathbf{s}^H(t)] > 0$
- $\mathbf{A} = [\mathbf{a}(\omega_1); \mathbf{a}(\omega_2); \dots; \mathbf{a}(\omega_p)]$
 - $\mathbf{a}(\omega_k) = [1; e^{j\omega_k}; \dots; e^{j\omega_k(n-1)}]$
 - $\omega_k = 2\pi \frac{d}{\lambda} \sin(\theta_k)$
 - d is the distance between the sensors in the antenna array. In this paper it will be equal the half of the wavelength λ .
 - θ_k is the direction of arrival of the k -th plane wave.
- $\mathbf{n}(t)$ is the noise vector.

The aim here, is to estimate and track the direction of arrivals $\{\theta_k\}$ in a robust way and with low computational cost. To do so, we first estimate adaptively a basis of the principal subspace of the observations covariance matrix $\mathbf{C}_x = E[\mathbf{x}\mathbf{x}^H]$ denoted $\mathbf{D}(t)$ using the HTFAPF robust algorithm. Then we proceed to the DoA extraction using the Total Least Squares version of ESPRIT (TLS-ESPRIT) as in [10]. Finally, we smooth the results using the Kalman filter to further enhance the estimation and tracking quality.

III. ROBUST SUBSPACE TRACKING ALGORITHM

In [10], Nguyen *et al* have proposed a new weighted cost function for subspace tracking. In this latter, a weighting parameter was introduced. Indeed, the observation is weighted so that corrupted observations are automatically mitigated.

$$J(\mathbf{D}(t)) = \sum_{i=1}^t \beta^{t-i} \omega(i) \|\mathbf{x}(i) - \mathbf{D}(t)\mathbf{y}(i)\|^2 \quad (2)$$

Where $0 < \beta < 1$ is a forgetting factor and $\mathbf{y}(i) = \mathbf{D}^H(t)\mathbf{x}(i) \approx \mathbf{D}^H(i-1)\mathbf{x}(i)$ according to the projection approximation introduced in [3].

The weight $\omega(i)$ is equal to the square inverse of Mahalanobis distance. It is straight forward to say that when this distance is large, proof of the presence of a corrupted observation, the weight becomes relatively very small.

$$\omega(i) = \frac{1}{d_M^2(\mathbf{x}(i), \mathbf{C}_x^{-1}(i-1))} = \frac{1}{\mathbf{x}^H(i)\mathbf{C}_x^{-1}(i-1)\mathbf{x}(i)} \quad (3)$$

To compute this distance with reduced cost, the inverse of the covariance matrix $\mathbf{C}_x^{-1}(t-1)$ is calculated recursively. Taking into account the instantaneous formula of the covariance matrix (4)

$$\mathbf{C}_x(t) = \beta \mathbf{C}_x(t-1) + \mathbf{x}(t)\mathbf{x}^H(t) \quad (4)$$

Later on, in [14], a more efficient cost function was proposed using FAPI's projection approximation introduced in [5] $\mathbf{D}(t) \approx \mathbf{D}(t-1)\Theta(t)$ where $\Theta(t)$ is an orthonormal matrix of size $p \times p$. This results in the cost function given by:

$$J'(\mathbf{D}(t)) = \sum_{j=1}^t \beta^{t-j} \omega(j) \|\mathbf{x}(j) - \mathbf{D}(t)\Theta(t)\mathbf{y}(j)\|^2 \quad (5)$$

Given that, two methods were then proposed to determine the weight value $\omega(t)$: the square inverse of Mahalanobis distance using the robust instantaneous formula of the covariance matrix (6) which leads to MFAPF algorithm:

$$\mathbf{C}_x(t) = \beta \mathbf{C}_x(t-1) + \omega(t)\mathbf{x}(t)\mathbf{x}^H(t) \quad (6)$$

and a lower-cost version using order statistics. The latter consists of using hard thresholding to determine whether the new weighting factor $\omega(t)$ takes 0 or 1 value at time instant t according to whether the observation is corrupted and disregarded, or is valid and included in the estimation of the subspace.

Indeed, considering the quantity $\epsilon^2(t)$ defined by :

$$\epsilon^2(t) = \|\mathbf{x}(t)\|^2 - \|\mathbf{y}(t)\|^2 \quad (7)$$

which represents approximately (up to a constant factor) the instantaneous noise power, and given the fact that impulsive noise realizations would have instantaneous power much higher and less frequent than the median power value, it was proposed in [14] to use the well known Inter Quartile Range (IQR) method for outliers detection [15], [16]. Using a sliding window of size L , we define:

$$\tilde{\epsilon}(t) = \{\epsilon^2(t-L+1) \dots \epsilon^2(t)\}$$

$$IQR(t) = Q_3(\tilde{\epsilon}(t)) - Q_1(\tilde{\epsilon}(t))$$

$$threshold(t) = Q_3(\tilde{\epsilon}(t)) + 1.5IQR(t) \quad (8)$$

Where $Q_1(\cdot)$ and $Q_3(\cdot)$ represent respectively the lower and the upper quartiles. Note that the adaptive computation of these quartiles is of negligible cost and many algorithms exist for its fast, adaptive evaluation, e.g. [17]. The weighting factor is then determined as follows:

$$\begin{cases} \omega(t) = 0 & \text{if } \epsilon^2(t) > threshold(t) \\ \omega(t) = 1 & \text{if } \epsilon^2(t) \leq threshold(t) \end{cases} \quad (9)$$

The above leads to the HTFAPF algorithm. Now, given the value of the weight factor, the resolution of the cost

Algorithm 1 HTFAPI algorithm

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1: Initialization
2:    $\mathbf{D}(0) = \begin{bmatrix} \mathbf{I}_p \\ 0_{(n-p) \times p} \end{bmatrix}; \mathbf{Z}(0) = \mathbf{I}_p$ 


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   for each time step do :
3:   input vector  $\mathbf{x}(t)$ 
4:    $\mathbf{y}(t) = \mathbf{D}^H(t-1)\mathbf{x}(t)$ 
5:    $\mathbf{h}(t) = \mathbf{Z}(t-1)\mathbf{y}(t)$ 
6:    $\epsilon^2(t) = \|\mathbf{x}(t)\|^2 - \|\mathbf{y}(t)\|^2$ 
   compute weight:
7:    $\tilde{\epsilon}(t) = [\epsilon^2(t-L+1) \cdots \epsilon^2(t)]$ 
8:    $IQR(t) = Q_3(\tilde{\epsilon}(t)) - Q_1(\tilde{\epsilon}(t))$ 
9:    $threshold(t) = Q_3(\tilde{\epsilon}(t)) + 1.5IQR(t)$ 
10:  if  $\epsilon^2(t) \leq threshold(t)$  then
11:     $\omega(t) = 1$ 
12:  else
13:     $\omega(t) = 0$ 
14:  end if
15:   $\mathbf{g}(t) = \frac{\mathbf{h}(t)\omega(t)}{\beta + \mathbf{y}^H(t)\mathbf{h}(t)\omega(t)}$ 
16:   $\tau(t) = \frac{\epsilon^2(t)}{1 + \epsilon^2(t)\|\mathbf{g}(t)\|^2 + \sqrt{1 + \epsilon^2(t)\|\mathbf{g}(t)\|^2}}$ 
17:   $\eta(t) = 1 - \tau(t)\|\mathbf{g}(t)\|^2$ 
18:   $\mathbf{y}'(t) = \eta(t)\mathbf{y}(t) + \tau(t)\mathbf{g}(t)$ 
19:   $\mathbf{h}'(t) = \mathbf{Z}^H(t-1)\mathbf{y}'(t)$ 
20:   $\boldsymbol{\epsilon}(t) = \frac{\tau(t)}{\eta(t)}(\mathbf{Z}(t-1)\mathbf{g}(t) - (\mathbf{h}'(t)^H\mathbf{g}(t))\mathbf{g}(t))$ 
21:   $\mathbf{Z}(t) = \frac{1}{\beta}(\mathbf{Z}(t-1) - \mathbf{g}(t)\mathbf{h}'(t)^H + \boldsymbol{\epsilon}(t)\mathbf{g}^H(t))$ 
22:   $\mathbf{e}'(t) = \eta(t)\mathbf{x}(t) - \mathbf{D}(t-1)\mathbf{y}'(t)$ 
23:   $\mathbf{D}(t) = \mathbf{D}(t-1) + \mathbf{e}'(t)\mathbf{g}^H(t)$ 

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function can be implemented as summarized in algorithm 1.

Once the principal subspace matrix $\mathbf{D}(t)$ is estimated in a robust way, we apply the TLS-ESPRIT algorithm [18] to estimate the directions of arrival at each time iteration. In what follows, we will use the Kalman filter to enhance the accuracy of those estimates.

IV. DOA SMOOTHING WITH KALMAN FILTER

In the following, we consider the case where the targets have time-varying directions. The state vector will therefore be $\mathbf{y}_k(i) = [\theta_k(i); \dot{\theta}_k(i)]$ where $\theta_k(i)$ is k^{th} direction of arrival and $\dot{\theta}_k(i)$ is its rate of variation. The dynamics of this motion, as well as the measurements, are modeled by equations (10) and (11):

$$\mathbf{y}_k(t+1) = \mathbf{F}\mathbf{y}_k(t) + \psi_k(t) \quad (10)$$

$$\theta_k(t) = \mathbf{H}\mathbf{y}_k(t) + v_k(t) \quad (11)$$

Where:

- $\mathbf{F} = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}$ is the transition matrix with a time step T , and $\mathbf{H} = [1 \ 0]$ is the measurement matrix.
- $\psi_k(t)$ is the noise process supposed to be Gaussian with covariance

$$\mathbf{Q}_k(t) = \begin{bmatrix} \frac{T^4}{2} & \frac{T^3}{2} \\ \frac{T^3}{2} & T^2 \end{bmatrix} \sigma_{\psi_k}^2$$

- $v_k(t)$ is the measurement noise with variance

$$R_k(t) = \sigma_{v_k}^2$$

The Kalman filter proceeds in two steps: a prediction before the measurement and an update after it:

A. Prediction step:

From the data available at time t , we can make a prediction of the state vectors of the next time step, of their covariance matrices $\mathbf{P}_k$, as well as of the following measurements:

$$\hat{\mathbf{y}}_k(t+1|t) = \mathbf{F}\mathbf{y}_k(t|t) \quad (12)$$

$$\mathbf{P}_k(t+1|t) = \mathbf{F}\mathbf{P}_k(t|t)\mathbf{F}^T + \mathbf{Q}_k(t) \quad (13)$$

$$\hat{\theta}_k(t+1|t) = \mathbf{H}\hat{\mathbf{y}}_k(t+1|t) \quad (14)$$

- At this point, the measurements $\theta_k(t)$ are taken to be the directions of arrival estimated by the HTFAPI algorithm followed by TLS-ESPRIT.

B. Update step:

After obtaining the new measurements, the prediction error, called innovation, and its covariance are computed as follows:

$$d_k(t) = \theta_k(t) - \hat{\theta}_k(t+1|t) \quad (15)$$

$$\mu_k = \mathbf{H}\mathbf{P}_k(t+1|t)\mathbf{H}^H + R_k(t) \quad (16)$$

This innovation is then used to improve the DoA estimation and their corresponding covariance updates:

$$\hat{\mathbf{y}}_k(t+1|t+1) = \hat{\mathbf{y}}_k(t+1|t) + \mathbf{G}_k(t+1)d_k(t) \quad (17)$$

$$\mathbf{P}_k(t+1|t+1) = [\mathbf{I}_2 - \mathbf{G}_k(t+1)\mathbf{H}] \mathbf{P}_k(t+1|t) \quad (18)$$

Where $\mathbf{I}_2$ is a 2×2 identity matrix and $\mathbf{G}_k(t+1)$ is the Kalman filter gain given by:

$$\mathbf{G}_k(t+1) = \mathbf{P}_k(t+1|t)\mathbf{H}^H\mu_k^{-1} \quad (19)$$

To initialize the Kalman filter we use the two points technique so that:

$$\mathbf{y}_k(2) = \left[\theta_k(2); \frac{\theta_k(2) - \theta_k(1)}{T} \right] \quad (20)$$

Therefor, the initial covariance depends on the measurement variance as follows

$$\mathbf{P}_k(2) = \begin{bmatrix} 1 & \frac{1}{T} \\ \frac{1}{T} & \frac{1}{T^2} \end{bmatrix} R_k(2) \quad (21)$$

Now, the choice of the Kalman filter parameters depends on the chosen model and the measurement accuracy.

To detect the presence of impulsive noise and thus to better tune the filter's parameters, we propose to use the weight $\omega(t)$ calculated in the HTFAPI algorithm. Indeed, since it takes two distinct values (0 or 1) according to whether there is a noise impulsion or not, we can determine the Kalman filter parameters according to the context as explained next.

V. SIMULATIONS AND RESULTS

In this section, we investigate the performance of our proposed algorithm according to the scenario below.

For the impulsive noise model, we consider here a burst noise generated according to equation (22):

$$\mathbf{n}(t) = \mathbf{n}_G(t) + \sum_{j=1}^{N_I} u\left(\frac{t - i_j}{b_j}\right) \mathbf{n}_I^j(t) \quad (22)$$

where $\mathbf{n}_G(t)$ is a white Gaussian and centered noise vector of variance σ_G^2 and $\mathbf{n}_I$ is a white-centered Gaussian noise of variance $\sigma_I^2 \gg \sigma_G^2$ weighted by $u(\cdot)$ a rectangular function, which is used to describe the short duration appearance of the burst noise. N_I refers to the number of impulsive events, i_j is the center of the j -th impulsive event, and b_j is its duration.

We aim to estimate and track the directions of arrival of $p = 3$ sources impinging on an antenna array of $n = 9$ sensors according to (1). We run 100 Monte Carlo simulations during $N = 1000$ instant time each.

The first source changes direction uniformly from 20° to 40° during the observation sequence, the second begins at 10° and ends the sequence at 20° while the third begins at 0° and ends at 5° , as displayed in Fig.(1):

The noise model followed is the one described in (22), where the signal is disturbed by an additive noise letting its signal-to-noise ratio (SNR) to be at $SNR = 0dB$ during the entire observation sequence except for two short periods of time: $P1 = [201, 250]$ and $P2 = [601, 700]$, where an impulsive noise appears inducing the signal SNR to be $SNR = -20dB$. A second simulation is run to show the behavior of the filters in the presence of a more important impulsive noise. In this latter, the signal SNR during $P1$ and $P2$ is set to be $SNR = -40dB$.

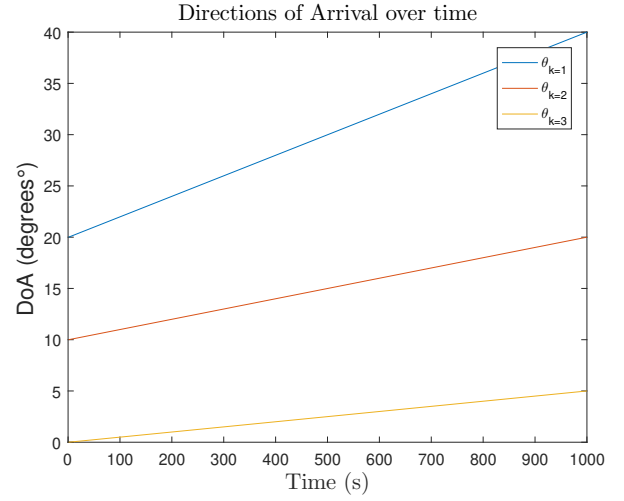


Fig. 1. Directions of arrival changes over the studied time sequence

The forgetting factor β in all algorithms including the HTFAPI is set to be 0.98 and the sliding window of the IQR method is of length $L = 1000$.

The Kalman filter parameters are adjusted as follows: the noise process is set to be $\sigma_{\psi_k}^2 = 0.00001^2$ during the entire experiment while the measurement noise is chosen dynamically as:

$$\begin{cases} \sigma_{v_k}^2 = 0.002 & \text{if } \omega(t) = 1 \\ \sigma_{v_k}^2 = 120 * 0.002 & \text{if } \omega(t) = 0 \end{cases}$$

The performance of this algorithm is evaluated according to the RMSE of the angular velocities $\omega_k = \pi \sin \theta_k$. The results shown in Fig.2 are compared to those of the OPASt algorithm as a benchmark to showcase the need for robust algorithms. Also, they are compared to those of the ROBust Subspace Tracking Algorithm (ROBUSTA) proposed in [10] and the RPAST [11] to illustrate the motivation behind the choice of the subspace tracking algorithm. Lastly, our algorithm is compared to the Kalman Filter with Variable Number of Measurements algorithm (KFVNM) [19] to showcase the performance difference as compared to a similar algorithm. It is clearly noticeable that our algorithm outperforms the other ones in the presence of impulsive noise. Moreover, this performance gain is more noteworthy when the impulsive noise is more important as shown in fig.3, when the signal SNR during the impulsion is about $SNR = -40dB$.

VI. CONCLUSION

We have proposed in this paper a new approach to enhance the directions of arrival estimation and tracking in an impulsive noise environment using a robust

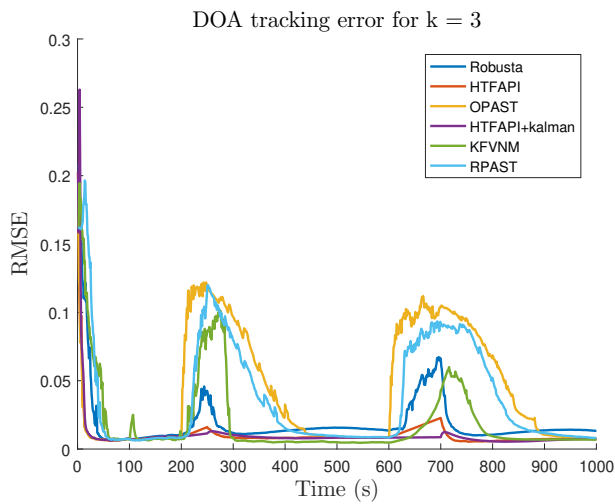
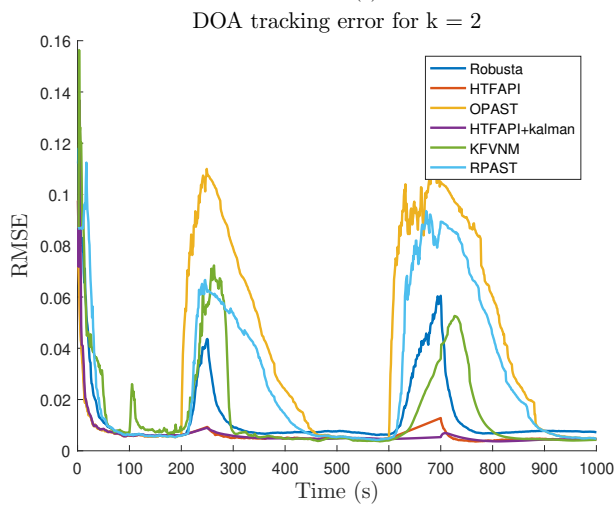
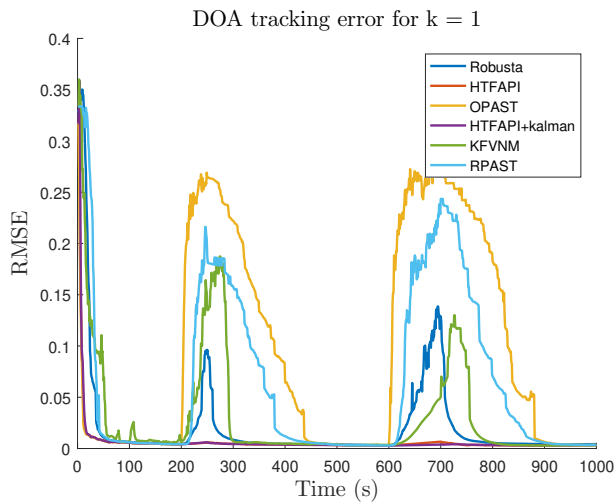


Fig. 2. RMSE of the DoA tracking in signal with SNR=-20dB in the presence of impulsive noise

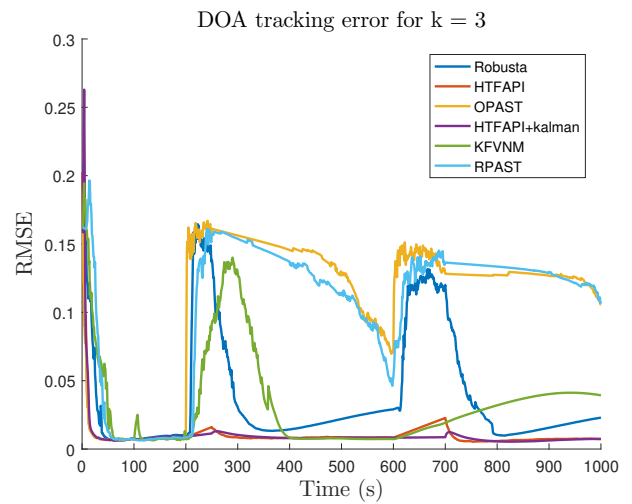
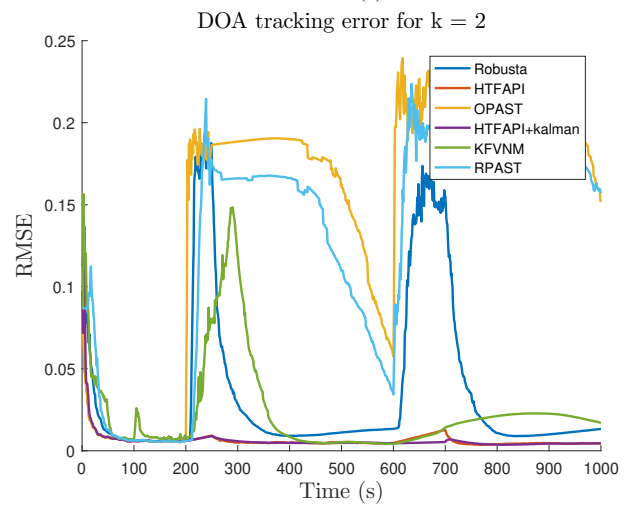
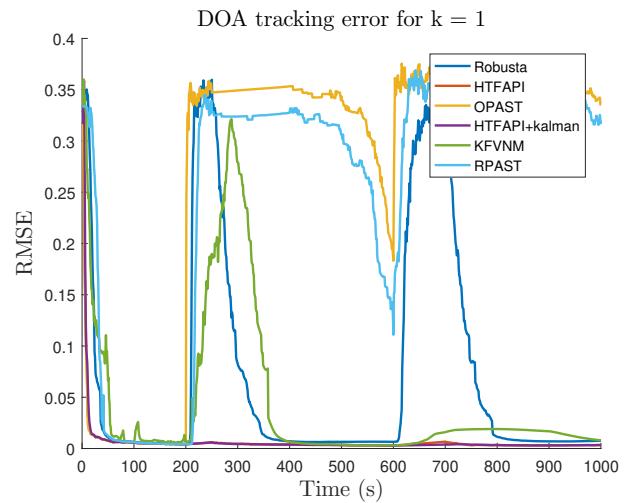


Fig. 3. RMSE of the DoA tracking in signal with SNR=-40dB in the presence of impulsive noise

subspace tracking method followed by a conventional Kalman filter. Indeed, the simulated experiments have shown that this latter combination improves considerably the performances of the tracking in the presence of a strong impulsive noise.

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